

TU VO

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Seoul, South Korea

PROFESSIONAL SUMMARY

Machine Learning Research Scientist with 5+ years of experience in Computer Vision and TinyML. Proven track record of publishing in top-tier conferences (WACV, ICCV) while engineering production-grade systems for microchips and cloud environments. Expertise in bridging the gap between theoretical research and deployable AI solutions.

TECHNICAL STRENGTHS

Languages	Python, MATLAB, SQL, Bash/Shell
Frameworks	PyTorch, TensorFlow, Scikit-learn, Pandas, LangChain, LangGraph
Research Focus	Computer Vision (Restoration, Deblurring), TinyML, Efficient Deep Learning

PUBLICATIONS

Drift-Augmented Scoring: Text-Derived Noise Robustness for Zero-Shot Audio-Language Classification

Tu Vo, Sheir Zaheer, Chan Y. Park

Accepted to 2026 ICML Workshop for Audio.

Error-Aware Deep Unrolling Network for Single Image Defocus Deblurring

Tu Vo, Chan Y. Park

Accepted to 2025 NeurIPS COML Workshop.

Deep Joint Unrolling for Deblurring and Low-Light Image Enhancement (JUDE)

Tu Vo, Chan Y. Park

WACV, 2025 — Project Page — Paper

Benchmarking Model Training and Inference for Resource-Constrained Deep Learning

... **Tu Vo** ...

ICCVW, 2023 — Paper

Attention! Stay Focus!

Tu Vo

CVPRW, 2021 — Paper — Code

High Dynamic Range Video Synthesis Using Superpixel-Based Illuminance-Invariant Motion Estimation

Tu Vo, Chul Lee

IEEE Access, 2023 — Paper

Robust HDR Video Synthesis Using Superpixel-Based Illumination Invariant Motion Estimation

Tu Vo, Chul Lee

IEEE ICCE-Asia, 2018 — Paper

Human Visual System Model-Based Optimized Tone Mapping of HDR Images

Nam Hoang Nguyen, **Tu Vo**, Chul Lee

IEEE Access, 2021 — Paper

EXPERIENCE

Research Scientist

KC Machine Learning Lab

March 2022 - Present

Seoul, South Korea

· **Image Restoration Research (WACV 2025, NeurIPS):** Spearheaded research on single-image defocus deblurring, resulting in a novel Deep Unrolling Network accepted to **WACV 2025, NeurIPS 2025 COML Workshop**.

- **Resource Constrained Vision (ICCVW 2023):** Developed a resource-efficient training pipeline for constrained environments. Authored reports and contributed to the "Resource Constrained Deep Learning" workshop published in ICCV 2023.
- **Efficient Super-Resolution (CVPR 2022 Challenge):** Researched and engineered lightweight models for image super-resolution tasks, conducting extensive benchmarking for the CVPR 2022 Efficient Super-Resolution Challenge.

Machine Learning Engineer

InhandPlus Inc.

March 2020 - January 2022

Seoul, South Korea

- **TinyML Sound Detection:** Engineered a lightweight sound detection model for the InhandPlus smart-watch, achieving **97% accuracy** in real-world trials. Optimized the model for low-power microchip deployment to trigger camera activation.
- **Behavior Analysis Pipeline:** Led the development of a patient monitoring system combining object detection (TensorFlow) and action recognition (LSTM). Built the training pipeline on **Azure Kubeflow** for daily model updates, achieving 95% accuracy in hospital deployments.
- **Auto-Labeling Tooling:** Developed an automated labeling tool (based on LabelImg) integrated with custom object detection models. Reduced manual data labeling time by **3x**, significantly accelerating dataset creation.

AI Engineer

Tricubics Inc.

February 2019 - March 2020

Seoul, South Korea

- **Smart Retail Object Detection:** Deployed a YOLOv3-based object detection system for smart retail environments. Integrated vision outputs with weight sensors for real-time inventory tracking at Emart Headquarters (90% system accuracy).
- **Synthetic Data Generation:** Developed an automated Data Generator using foreground segmentation to synthesize training data with randomized backgrounds, solving data scarcity issues for retail items.

PERSONAL PROJECTS

Attention! Stay Focus! - CVPR Workshop 2021 ([link](#))

- A deep learning model to solve the Defocus Deblurring Problem. The deep learning model is modified from the original Unet with the Attention Mechanism. CVPRW 2021
- Github link: ATTSF

Zero-DCE TF:

A TensorFlow implementation of Zero-DCE, a deep learning-based image enhancement algorithm that functions without paired input-ground truth data.

Link: Zero_DCE_TF

EDUCATION

Pukyong National University, South Korea

MS in Computer Engineering

February 2019

Hanoi University of Science and Technology, Vietnam

Bachelor in Electrical & Electronic Engineering (Advanced Program)

July 2016

REFERENCES

Prof. Chul Lee

Dongguk University

Seoul, South Korea

- Email: chullee@dongguk.edu
- Relationship: Master Degree Supervisor

Prof. Hoai Linh Tran

Hanoi University of Science and Technology

Hanoi, Vietnam

- Email: linh.tranhoai@hust.edu.vn
- Relationship: Bachelor Degree Advisor